

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Code of Conduct:

I S. Nithesh (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Nithesh

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%				
Debugging	25%				
Optimization	20%				
Analysis	20%				
Code Quality	15%				

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

IP-4 Debugging & Optimization:

IP Address : 192.168.10.65

Subnet Mask : 255.255.255.192 (/26)

Default Gateway : 192.168.10.1

Step 1: Identify the Subnet

A/26 subnet mask means each subnet contains 64 IP addresses.

Subnet ranges are:

192.168.10.0 — 192.168.10.63

192.168.10.64 — 192.168.10.127

192.168.10.128 — 192.168.10.191

192.168.10.192 — 192.168.10.255

Step 2: Network Details:

Network Address : 192.168.10.64

Broadcast Address : 192.168.10.127

First Host : 192.168.10.65

Last Host : 192.168.10.126

Step 3: Verify the Configuration

The IP address is valid because it lies within the host range. However, the default gateway 192.168.10.1 belongs to 192.168.10.0/26 subnet, which is different from the workstation subnet.

Step 4: Configuration Error:

The default gateway is configured in the wrong subnet.

Step 5: Optimized Configuration:

IP Address: 192.168.10.65

Subnet Mask: 255.255.255.192

Default Gateway: 192.168.10.126

Step 6: Number of Usable Hosts:

for a /26 subnet

Total addresses = 64

Network address = 1

Broadcast address = 1

Usable hosts = 62

IP v6 Debugging:
Device A: 2001:0000:0000:0000:0000:0000:0000:0000
Device B: 2001:0000:0000:0000:0000:0000:0000:0000

Step 1: 1
Device A: 2001:0000:0000:0000:0000:0000:0000:0000
Device B: 2001:0000:0000:0000:0000:0000:0000:0000

IPv6 Debugging:

Device A : 2001::db8::ff00:42:8329 /64

Device B : 2001::db8:0:0:1::1 /64

Step 1: Expand IPv6 Addresses:

Device A:

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B:

2001:0db8:0000:0000:0001:0000:0000:0001

Step 2: Determine the Network Prefix

Since both use a /64 prefix, the first 64 bits represent the network.

Network Prefix : 2001::db8:0:0:: /64

Step 3: Verify Subnet

Both devices belongs to the same /64 network.

Step 4: Identify the Problem:

The IPv6 addresses are correctly assigned.

∴ communication failure is likely caused by:

- * Incorrect IPv6 gateway.
- * Disabled IPv6 interface.
- * Router Configuration Problem
- * Firewall restriction.

Step 5: Optimized Configuration.

Device A:

2001:db8:0:0:10/64

Device B:

2001:db8:0:0:20/64

Step 6: Number of Host Addresses:

A/64 subnet provides: $2^{24} =$

18,446,744,073,709,551,616 addresses.

IP v4 Subnet 0
Given:-
Network
Subnet
* /26
* /

IP v4 Subnet Optimization using VLSM

Given:-

Network = 192.168.1.0/24

Subnets:

* /26

* /27

* /28

Subnet 1 (/26):

* Network Address : 192.168.1.0

* First Host : 192.168.1.1

* Last Host : 192.168.1.62

* Broadcast : 192.168.1.63

* Total IPs : 64

* Usable Hosts : 62

Subnet 3 (/28):

Network Address : 192.168.1.96

First Host : 192.168.1.97

Last Host : 192.168.1.110

Broadcast : 192.168.1.111

Total IPs : 16

Usable Hosts : 14

Subnet 2 (127):

Network Address: 192.168.1.64

First Host: 192.168.1.65

Last Host: 192.168.1.94

Broadcast: 192.168.1.95

Total IPs: 32

Usable Hosts: 30

Analysis:

Some departments receive more IP addresses than required, while others receive fewer addresses.

Debugging:

The addressing plan should be checked to ensure each department receives the correct subnet according to its size.

Optimization Using VLSM:

Allocate larger subnet to department with more users & smaller subnet to department with fewer users.

192.168.1.0/24 Debugging:

Expander Addresses:

Device A:

2001:0db8:

Device B:

2001:0db8:

Network

Sub

pr

IPv6 Debugging:

Expanded Addresses:

Device A:

2001:0db8:0000:0000:0000:ff00:0042:8329.

Device B:

2001:0db8:0000:0000:0001:0000:0000:0001

Network Prefix 2001:db8:0:0::/64

Subnet Verification for both devices are in the same subnet.

Problem:

The issue is likely related to IPv6 routing, gateway configuration, interface status, or firewall settings.

Optimized Addresses:

Device A:

2001:db8:0:0::10/64

Device B:

2001:db8:0:0::20/64

Available Host Addresses

$2^{64} = 18,446,744,073,709,551,616$ addresses.

Conclusion:

The IPv6 addressing is correct. The communication problem should be investigated at the router or interface configuration level.

Routing Table Debugging.

Routing Table:

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Destination IP:

192.168.2.100

Step 1: Longest Prefix Match

The destination IP belongs to 192.168.2.0/24, so this is the matching route.

Step 2: Next - Hop Router

The router forwards the packet to 10.0.0.3.

Step 3: Verify Destination

The IP address 192.168.2.100 falls within the range 192.168.2.1 — 192.168.2.254, so it belongs to the correct subnet.

Step 4: Possible Routing Errors

- * Incorrect next-hop address
- * Router interface is down
- * Missing route in the routing table.
- * Physical cable or link failure.

Step 5: Default Route

If no destination network matches, the router forwards packets to the default gateway (10.0.0.1).